



**Cambridge Instruments
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

OIL TEST SET · TRANSFORMER OIL BDV TESTER

CIE/7000 SERIES

Transformer oil dielectric strength (BDV) tester series - 0-50 kV to 0-120 kV, 8 models. Manual and motorised variants.

50-120kV

Transformer Oil BDV Tester Series

A range of oil dielectric strength test sets for breakdown/withstand voltage testing of transformer and circuit breaker insulating oil per IS 6792:1992, spanning eight output-voltage tiers from 0-50kV (CIE/7050) up to 0-120kV (CIE/7120). Every model shares the same design and is available in manual and motorised (MOT) drive.

- **8 model tiers** - 0-50kV to 0-120kV, one model per rating
- **Manual or motorised** - motorised advances at approx. 2kV/second
- **Memory device** - retains breakdown reading after tripping
- **Zero-start interlocking** - plus interlocked H.T. door cutout
- **Analog moving-iron metering** - 96mm² kV meter, L.T. side
- **Standard test vessel** - brass spheres, per IS 6792:1992



- V **50-120kV** 8 distinct model tiers
- M **Manual/Motorised** Approx. 2kV/second (motorised)
- ✓ **Memory Device** Retains breakdown reading

Manufactured in India since 1963 - trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors.

01 Full Specification Sheet

PARAMETER	SPECIFICATION
Output Voltage Range	0-50 kV to 0-120 kV (8 models, continuously variable)
Capacity Range	500 VA to 1200 VA (varies by model)
Input	230 V $\pm$ 10%, 50 Hz, 1 phase A.C.
Duty	Intermittent
H.V. Winding	Cast-resin, graded insulation, centre point grounded
Metering	96 mm ² A.C. moving-iron voltmeter, L.T. side, scaled in kV (analog)
Motorised Rate	Approx. 2 kV/second (motorised variant)
Test Vessel	Two brass spheres, 12.5-13 mm dia., 4mm or 2.5mm gap, per IS 6792:1992
Warranty	12 months against manufacturing defects

Any requirement apart from the standard voltage tiers & drive above can be built to order.

02 CIE/7000 Series - Output Voltage & Capacity

OUTPUT VOLTAGE	CAPACITY	MANUAL MODEL	MOTORISED MODEL
0-50 kV	500 VA	CIE/7050	CIE/7050 MOT
0-60 kV	600 VA	CIE/7060	CIE/7060 MOT
0-70 kV	700 VA	CIE/7070	CIE/7070 MOT
0-75 kV	750 VA	CIE/7075	CIE/7075 MOT
0-80 kV	800 VA	CIE/7080	CIE/7080 MOT
0-90 kV	900 VA	CIE/7090	CIE/7090 MOT
0-100 kV	1000 VA	CIE/7100	CIE/7100 MOT
0-120 kV	1200 VA	CIE/7120	CIE/7120 MOT

03 Typical Applications

- Transformer oil dielectric strength testing before energisation
- Routine oil quality monitoring for distribution and power transformers
- Circuit breaker oil testing
- Power utility and substation maintenance laboratories

04 What's in the Box

- ✓ Instruction manual with circuit diagram
- ✓ Input supply cable, minimum 2 metres, 3-pin plug
- ✓ Test vessel (oil cell) with electrode and gap-setting gauge

05 Optional Accessories

- + Alternate tripping current set point
- + Digital metering in place of analog
- + 2-unit housing
- + Double capacity, 20 mA (CIE/7050-7080 only)
- + Magnetic stirrer